Model Answer and Marking Scheme

Internal Assessment 1 - Engineering Physics | Maximum marks: 20

Q1. Newton's laws of motion [5]

- First law (inertia): a body remains at rest or in uniform motion in a straight line unless acted on by an external unbalanced force. [1.5]
- Second law: the rate of change of momentum is proportional to the applied force and takes place in its direction; F = ma. [1.5]
- Third law: to every action there is an equal and opposite reaction (forces act on different bodies). [1]
- Example of third law, e.g. recoil of a gun, rocket propulsion, swimmer pushing water backwards. [1]

Q2. Uniformly accelerated motion [5]

(a) u = 0, v = 20 m/s, t = 5 s. a = (v - u) / t = (20 - 0) / 5 = 4 m/s^2. [formula 1, answer with unit 1]
(b) s = ut + (1/2)at^2 = 0 + 0.5 x 4 x 25 = 50 m. [formula 1, substitution 1, answer with unit 1]
(Alternative: s = average velocity x t = 10 x 5 = 50 m, full marks.)
Common error: s = v x t = 100 m - award at most 1 mark.

Q3. Ohm's law [5]

- Statement: at constant temperature (and other physical conditions), the current through a conductor is directly proportional to the potential difference across it; V = IR. [2]
- Graph: straight line through the origin, axes labelled V and I with units. [2]
- Resistance = slope of the V-I graph (V on y-axis, I on x-axis), R = V / I = constant. [1]

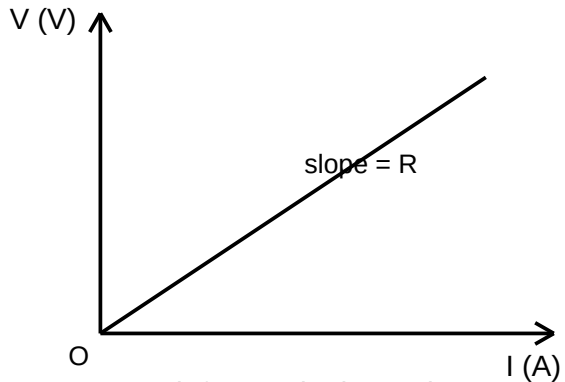

Figure: V-I graph for an ohmic conductor - straight line through origin O; slope gives R.

Q4. Energy [5]

- Kinetic energy: energy possessed by a body due to its motion, KE = (1/2)mv^2. [1]
- Potential energy: energy possessed by a body due to its position/configuration, PE = mgh. [1]
- Energy conservation: loss in PE = gain in KE, mgh = (1/2)mv^2, so v = sqrt(2gh). [1.5]
- v = sqrt(2 x 10 x 20) = sqrt(400) = 20 m/s. [1.5]
(Using v^2 = u^2 + 2gh also gets full marks.)